Music Genre Classifier

A Machine Learning Pipeline for Audio Classification

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1. OVERVIEW

This project builds an end-to-end audio machine learning pipeline to automatically classify music into genres using low-level audio features. It was built as a personal introduction to Music Information Retrieval (MIR), exploring how well computational methods can capture the acoustic characteristics that define a musical genre.

The project also investigates a deeper question: what happens when the training data does not represent the full diversity of the world's music? The GTZAN dataset used here is composed almost entirely of Western genres; no Afrobeat, no Mbalax, no Coupé-Décalé. A model trained on this data would likely fail on music from traditions it has never seen. This cultural blind spot is a known open problem in MIR research and one of the core motivations behind this work.

2. DATASET

Dataset: GTZAN Music Genre Dataset

- 1,000 audio files, 30 seconds each, across 10 genres
- Genres: blues, classical, country, disco, hiphop, jazz, metal, pop, reggae, rock
- 100 tracks per genre
- Note: jazz.00054.wav is a known corrupted file in the dataset ;skipped automatically with try/except
- 999 files successfully processed

3. STEP 1 - FEATURE EXTRACTION

Each audio file is loaded using librosa with a duration of 30 seconds:

```
y, sr = librosa.load(file_path, duration=30)
```

y is the audio signal; an array of 661,500 numbers where each number represents the amplitude of the sound at a given instant. sr is the sample rate (22,050 Hz by default), meaning 22,050 samples are captured per second.

librosa slices the signal into small overlapping windows. By default, the window advances by 512 samples each time (hop_length). This gives:

```
number of frames = 661,500 / 512 ≈ 1,292 frames
```

For each frame, a set of descriptors is computed. Since ML models require a fixed-size input vector per song, all frame-level features are summarized using their mean (and standard deviation where applicable).

Features Extracted

Feature	Description	Dimensions

MFCCs (mean + std)	Timbral texture — how the sound 'sounds'	26
Chroma	Harmonic content; which notes are played	12
Spectral Contrast	Difference between peaks and valleys in spectrum	7
Tempo	BPM; rhythmic speed (single scalar, no mean needed)	1
Zero Crossing Rate	How often the signal crosses zero (percussiveness)	1
Spectral Rolloff	Frequency below which 85% of energy falls	1
Spectral Centroid	Brightness of the sound	1
Spectral Bandwidth	Width of the spectral distribution	1

Total feature vector per song: 50 dimensions

4. STEP 2 - PREPROCESSING

Before training, two preprocessing steps are applied:

Label Encoding

Genre labels are strings (e.g. 'blues', 'jazz') but ML models require numerical inputs. LabelEncoder converts them to integers: blues → 0, classical → 1, country → 2, etc.

Feature Scaling

The 50 features have very different scales; tempo is around 120 BPM while MFCCs are small decimals. Without scaling, the model would give disproportionate weight to features with larger values. StandardScaler brings everything to the same scale (mean=0, std=1).

Golden rule applied throughout:

- Train set: fit_transform(); the scaler learns the mean and std from training data, then applies it
- Test set: transform() only; the scaler applies the same transformation learned from training data, never re-fits

5. STEP 3 - TRAIN / TEST SPLIT

The dataset is split 80/20 using train_test_split with stratify=y to ensure each genre is proportionally represented in both sets:

- Training set: ~799 files
- Test set: ~200 files (20 per genre)

6. STEP 4 - MODELS

SVM : Support Vector Machine

The SVM finds an optimal boundary (hyperplane) that separates genres in the 50-dimensional feature space. Key parameters:

- kernel='rbf' (Radial Basis Function); allows the model to draw complex, non-linear boundaries between genre
- C=10; controls the importance of errors. High C means fewer training errors but risks overfitting

- gamma='scale'; controls how much influence each point has. 'scale' lets sklearn choose automatically
- random_state=42; ensures reproducible results

Neural Network : MLPClassifier

A fully connected multi-layer perceptron. Each layer learns increasingly abstract patterns from the audio features. Best configuration found via Grid Search:

- hidden_layer_sizes=(512, 256, 128) — three hidden layers
- alpha=0.0001 — regularization to prevent overfitting
- max_iter=500

7. STEP 5 - HYPERPARAMETER TUNING (GRID SEARCH + CROSS-VALIDATION)

Instead of manually choosing hyperparameters, GridSearchCV automatically tries all combinations and evaluates them using 5-fold cross-validation:

5-fold cross-validation splits the training set into 5 equal parts. In each round, 4 parts are used for training and 1 for validation, rotating until every part has served as validation once. The final score is the average of the 5 rounds. This gives a more reliable estimate than a single train/val split.

- SVM: tested 16 combinations (4 values of C × 2 gamma × 2 kernels) × 5 folds = 80 models
- Neural Network: tested 6 combinations × 5 folds = 30 models

Best SVM parameters found: C=10, gamma='scale', kernel='rbf'

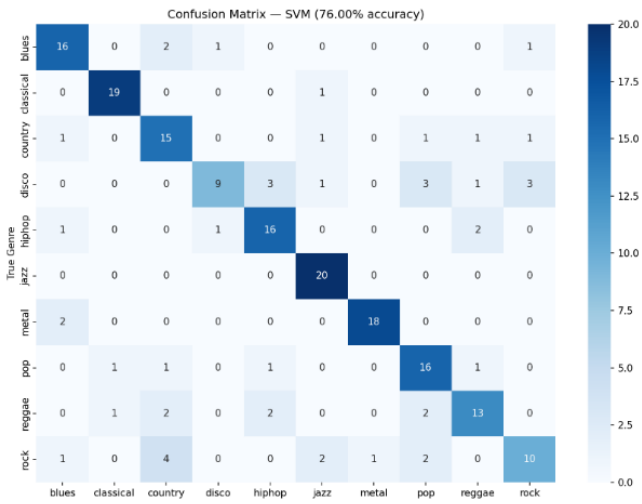
Best MLP parameters found: hidden_layer_sizes=(512, 256, 128), alpha=0.0001

8. RESULTS

- RESULTS1 : 3 features: timbre, tempo, harmonie

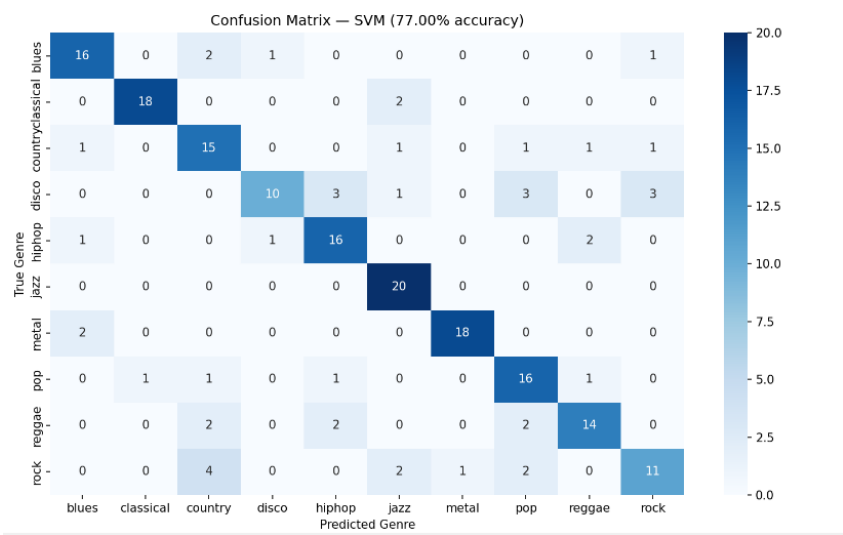
```
Classification Report:
              precision    recall  f1-score   support

       blues       0.61      0.55      0.58        20
   classical       0.89      0.85      0.87        20
     country       0.50      0.70      0.58        20
       disco       0.35      0.30      0.32        20
      hiphop       0.67      0.60      0.63        20
        jazz       0.68      0.95      0.79        20
       metal       0.88      0.75      0.81        20
         pop       0.68      0.75      0.71        20
      reggae       0.61      0.55      0.58        20
        rock       0.47      0.35      0.40        20

    accuracy                           0.64       200
   macro avg       0.63      0.63      0.63       200
weighted avg       0.63      0.64      0.63       200

Confusion Matrix:
[[11  0  3  1  0  2  0  0  2  1]
 [ 0 17  0  0  0  3  0  0  0  0]
 [ 1  0 14  0  0  2  0  1  1  1]
 [ 1  0  1  6  3  1  0  3  1  4]
 [ 0  0  1  5 12  0  0  1  1  0]
 [ 1  0  0  0  0 19  0  0  0  0]
 [ 3  0  0  1  0  0 15  0  0  1]
 [ 0  1  0  1  2  0  0 15  1  0]
 [ 0  1  2  3  1  0  0  1 11  1]
 [ 1  0  7  0  0  1  2  1  1  7]]
Accuracy: 0.635
```

- **RESUJLT 2: adding special contrast, zero crossing rate and spectral rolloff**



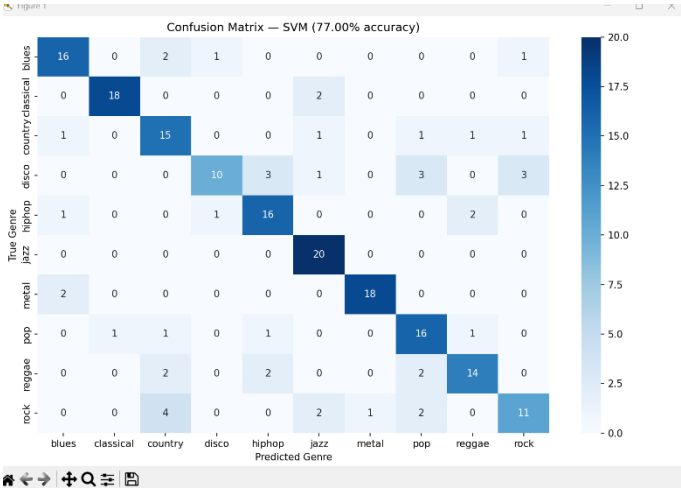

```
Skipped jazz.00054.wav:
SVM Accuracy: 76.00%

Training Neural Network...
Neural Net Accuracy: 72.50%

Best model: SVM (76.00%)

Classification Report:
              precision    recall  f1-score   support

       blues       0.76      0.80      0.78        20
   classical       0.90      0.95      0.93        20
     country       0.62      0.75      0.68        20
       disco       0.82      0.45      0.58        20
      hiphop       0.73      0.80      0.76        20
        jazz       0.80      1.00      0.89        20
       metal       0.95      0.90      0.92        20
         pop       0.67      0.80      0.73        20
      reggae       0.72      0.65      0.68        20
        rock       0.67      0.50      0.57        20

    accuracy                           0.76       200
   macro avg       0.76      0.76      0.75       200
weighted avg       0.76      0.76      0.75       200
```

- **RESULT 3:: adding spectral centroid and spectral brandwidth**




```
SVM Accuracy: 77.00%

Training Neural Network...
Neural Net Accuracy: 73.50%

Best model: SVM (77.00%)

Classification Report:
              precision    recall  f1-score   support

       blues       0.80      0.80      0.80        20
   classical       0.95      0.90      0.92        20
     country       0.62      0.75      0.68        20
       disco       0.83      0.50      0.62        20
      hiphop       0.73      0.80      0.76        20
        jazz       0.77      1.00      0.87        20
       metal       0.95      0.90      0.92        20
         pop       0.67      0.80      0.73        20
      reggae       0.78      0.70      0.74        20
        rock       0.69      0.55      0.61        20

    accuracy                           0.77       200
   macro avg       0.78      0.77      0.77       200
weighted avg       0.78      0.77      0.77       200
```

- **RESULT4: same features + tuned hyperparametres :**




```
Training SVM with Grid Search (this takes ~15 min)...
Fitting 5 folds for each of 16 candidates, totalling 80 fits
Best SVM params: {'C': 10, 'gamma': 'scale', 'kernel': 'rbf'}
SVM Accuracy: 77.00%

Training Neural Network with Grid Search...
Fitting 5 folds for each of 6 candidates, totalling 30 fits
Best MLP params: {'alpha': 0.0001, 'hidden_layer_sizes': (512, 256, 128), 'max_iter': 500}
Neural Net Accuracy: 72.50%

Best model: SVM (77.00%)

Classification Report:
              precision    recall  f1-score   support

       blues       0.80      0.80      0.80        20
   classical       0.95      0.90      0.92        20
     country       0.62      0.75      0.68        20
       disco       0.83      0.50      0.62        20
      hiphop       0.73      0.80      0.76        20
        jazz       0.77      1.00      0.87        20
       metal       0.95      0.90      0.92        20
        pop       0.67      0.80      0.73        20
      reggae       0.78      0.70      0.74        20
        rock       0.69      0.55      0.61        20

    accuracy                           0.77       200
   macro avg       0.78      0.77      0.77       200
weighted avg       0.78      0.77      0.77       200
```

Progression across experiments Recap

Experiment	Features Added	SVM Accuracy
Result 1	MFCCs + Chroma + Tempo	63.5%
Result 2	+ Spectral Contrast + ZCR + Rolloff	76.0%
Result 3	+ Spectral Centroid + Bandwidth	77.0%
Result 4 (Grid Search)	Same features + tuned hyperparams	77.0%

Adding more features consistently improved accuracy. The biggest jump came from Result 1 to Result 2, confirming that spectral contrast and zero crossing rate carry significant genre-discriminative information.

Best model: SVM (77.0%)

Genre	Precision	Recall
blues	0.80	0.80
classical	0.95	0.90
country	0.62	0.75
disco	0.83	0.50
hiphop	0.73	0.80
jazz	0.77	1.00
metal	0.95	0.90
pop	0.67	0.80

reggae	0.78	0.70
rock	0.69	0.55

Key observations:

- Jazz: perfect recall (1.00); the model never misses a jazz track
- Classical & Metal: precision ~95%;when the model says classical or metal, it's almost always right
- Rock & Disco: weakest performance; rhythmically and tonally similar to several other genres
- Country: often confused with rock and pop, which share similar tempos and chord progressions

9. REFLECTIONS & LIMITATIONS

On cultural representation

GTZAN contains exclusively Western popular music genres. A model trained on this dataset has no exposure to West African rhythmic traditions; Afrobeat, Highlife, Coupé-Décalé, Mbalax, or to music from the Middle East, South Asia, or other non-Western traditions. This is not just a data gap; it reflects a structural bias in how MIR benchmarks have historically been constructed.

Coming from Côte d'Ivoire, this limitation is personally significant. The question of whether computational systems can understand the rhythmic complexity of Ivorian or West African music, traditions with fundamentally different rhythmic structures than Western pop, is an open research problem. This connects directly to findings by Bello et al. (2015), who showed that even a technically well-performing rhythm similarity system remained culturally blind when its training data did not represent the music being analyzed.

On genre as a proxy

Genre labels are inherently fuzzy. Rock and country share rhythmic patterns; disco and pop overlap harmonically. High classification accuracy does not guarantee that the model truly understands musical similarity; it may be picking up on artifacts (like recording quality, as shown in GTZAN's Tango class) rather than musical content.

10. REFERENCES

- Tzanetakis & Cook (2002) -Musical Genre Classification of Audio Signals (original GTZAN paper)
- Esparza, Bello & Humphrey (2015) - From Genre Classification to Rhythm Similarity
- McFee et al. (2015) 0librosa: Audio and Music Signal Analysis in Python
- Bello et al. (2010) -Automated Music Emotion Recognition